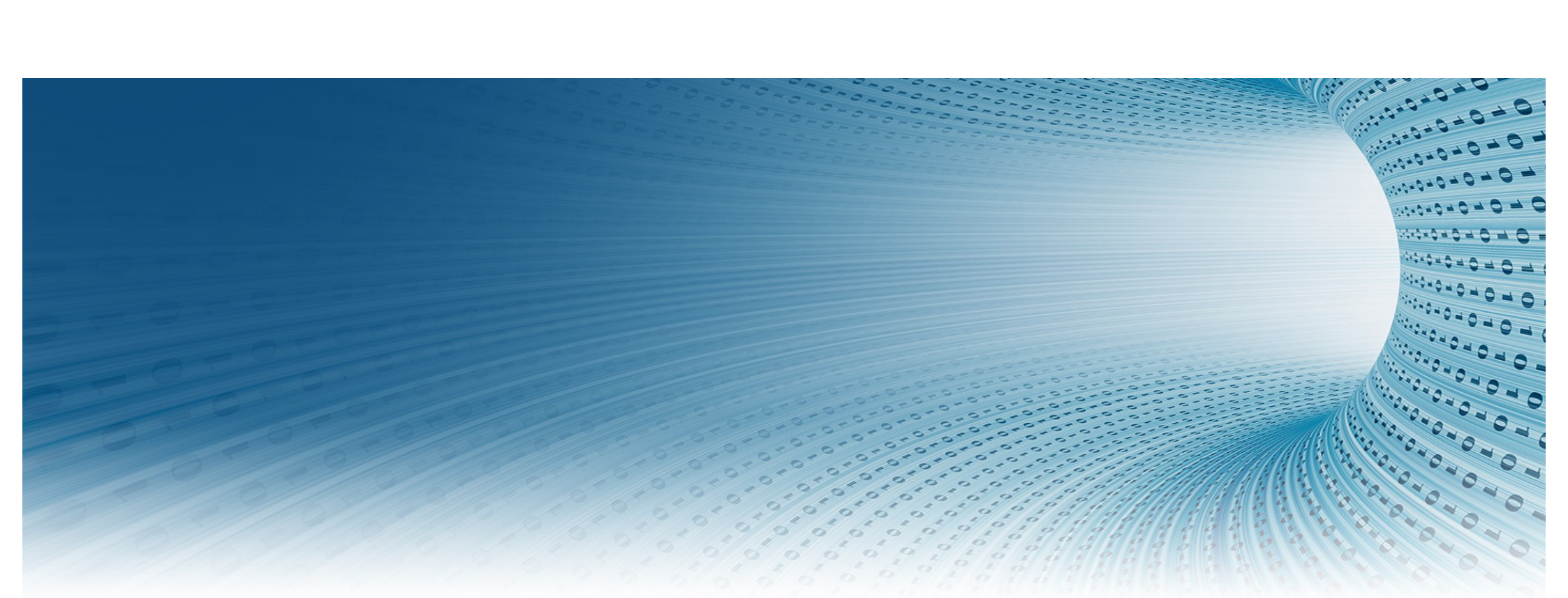




# Selecting Input Probability Distributions

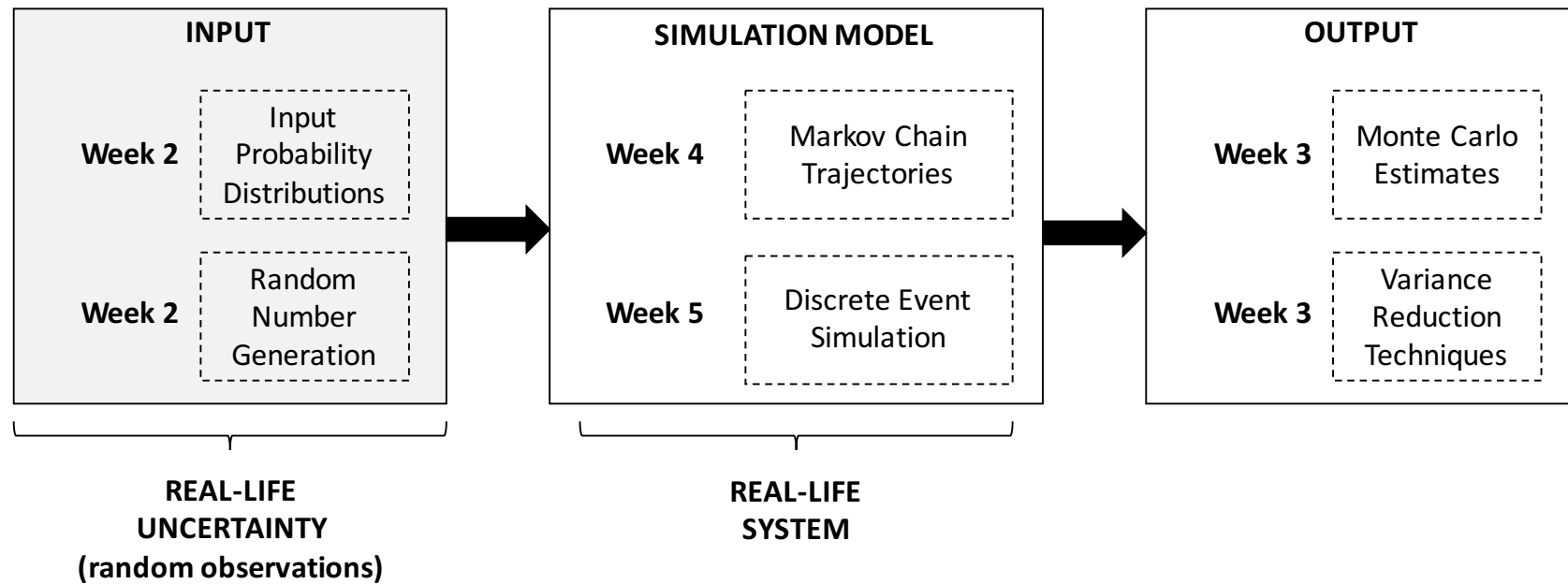
## Empirical and Theoretical Distributions

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# Course Schedule

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# Outline

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- Introduction
- Specification of empirical distributions
- Parametrisation of theoretical distributions
- Specification of theoretical distributions

# Introduction

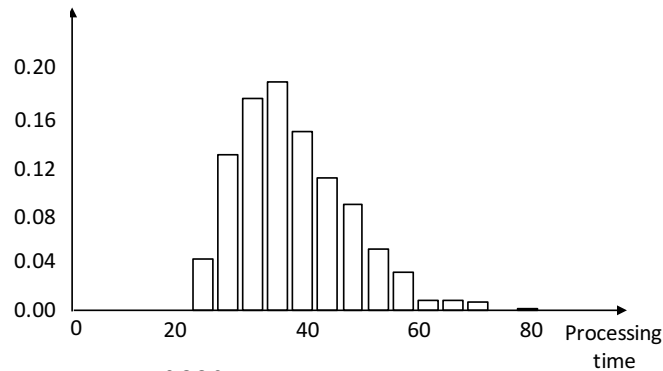
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- To carry out a simulation using random inputs, probability distributions need to be defined
- Almost all real-world systems contain one or more sources of randomness

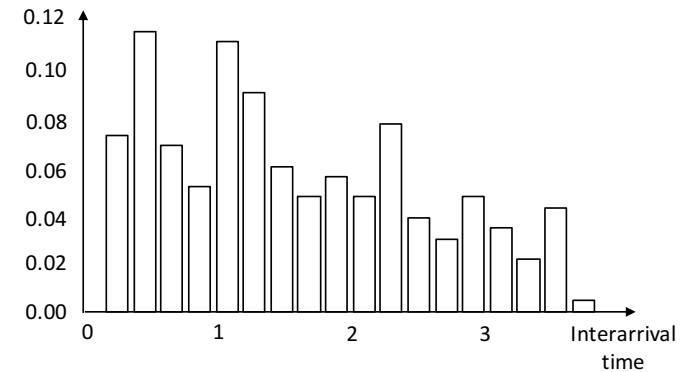
| Type of system  | Sources of randomness                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------------------------|
| Manufacturing   | Processing times, machine times to failure, machine repair times                                            |
| Defense-related | Arrival times and payloads of missiles or airplanes, outcome of an engagement, miss distances for munitions |
| Communications  | Interarrival times of messages, message types, message lengths                                              |
| Transportation  | Ship-loading times, interarrival times of customers to a subway                                             |

# Introduction

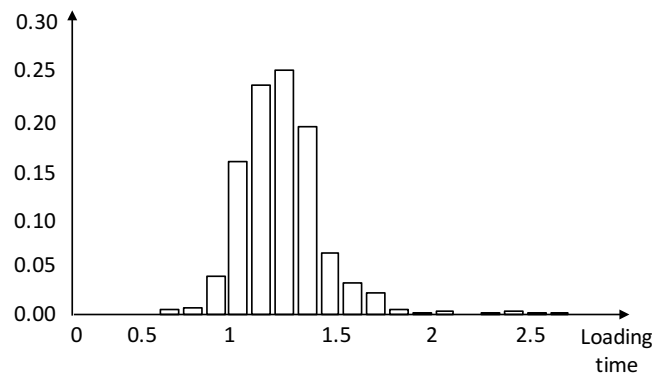
- Example histograms



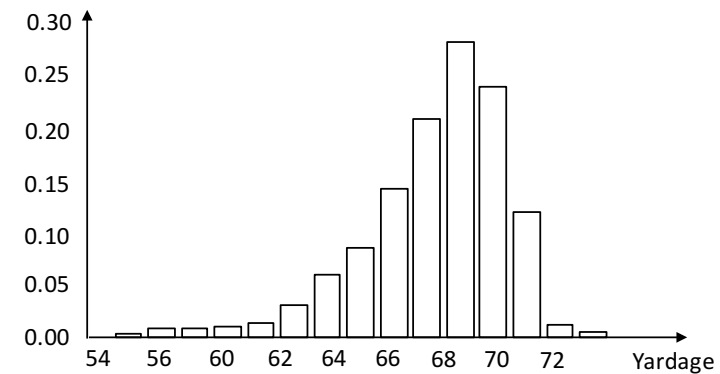
Histogram of 890 machine processing times



Histogram of 219 interarrival times to a drive-up bank



Histogram of 856 ship-loading times



Histograms of the yardages for 1000 large rolls of paper for a household-products manufacturer

# Introduction

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- Data of an input random variable can be used to
  1. The data values can be used directly in the simulation (*Trace-driven simulation*)
    - Advantage: Recommended for model validation
    - Disadvantages
      - Disadvantage 1: The simulation can only reproduce what has happened historically
      - Disadvantage 2: There is seldom enough data to make all the desired number of simulation runs
  2. The data values can be used to define an *empirical distribution* function
    - Empirical distributions are required when
      - The data is a mixture of two or more heterogeneous populations
      - There does not exist a good fit with a continuous theoretical distribution (e.g. the times to perform some task have been significantly rounded, theoretical distributions need to be truncated)

# Introduction

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- Data of an input random variable can be used to
  2. The data values can be used to define an *empirical distribution* function

(not in your slides)

- Example

Observations: 2 – 2 – 3 – 2 – 1

Empirical distribution:  $\Pr[X = 1] = 0.2$        $\Pr[X = 2] = 0.6$        $\Pr[X = 3] = 0.2$

In a trace-driven simulation      2 – 2 – 3 – 2 – 1      (exact replication of observations)

In a simulation experiment      2 – 1 – 2 – 2 – 2      (randomised sequence of observations)

with empirical distribution

# Introduction

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- Data of an input random variable can be used to
  3. Standard techniques of statistical inference are used to fit a *theoretical distribution* form to the data and perform hypothesis tests to determine the goodness of fit
    - Advantages of theoretical distribution compared to empirical distribution

Advantage 1: An empirical distribution may have certain irregularities (certainly if only a small number of data values are available). A theoretical distribution smooths out the data and provides information on overall distribution

Advantage 2: If empirical distributions are used in the usual way, it is not possible to generate values outside the range of observed data. This is possible with theoretical distributions.

Advantage 3: Compact way of representing a set of data values. Empirical distributions are cumbersome.

Advantage 4: A theoretical distribution is easier to change.



# Introduction

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- Data of an input random variable can be used to
  2. The data values can be used to define an *theoretical distribution* function

(not in your slides)

- Example

Observations: 2 – 2 – 3 – 2 – 1

Theoretical distribution:      Normal distribution  $\sim N(2,1)$

|                                                             |                   |                                       |
|-------------------------------------------------------------|-------------------|---------------------------------------|
| In a trace-driven simulation                                | 2 – 2 – 3 – 2 – 1 | (exact replication of observations)   |
| In a simulation experiment<br>with empirical distribution   | 2 – 1 – 2 – 2 – 2 | (randomised sequence of observations) |
| In a simulation experiment<br>with theoretical distribution | 0 – 2 – 2 – 3 – 4 | (randomised sequence of observations) |

# Empirical distributions

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- An empirical distribution is specified directly by the observed data

- Continuous random variables

- Case I: The individual original observations  $X_1, X_2, \dots, X_n$  are known

We can define a continuous piecewise-linear distribution  $F$  as follows

Step 1: Sort the individual original observations  $X_j$  into increasing order with

$X_{(i)}$  the  $i$ th smallest of the  $X_j$ 's, i.e.  $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$

Step 2: Define the cumulative density distribution  $F(x)$

$$F(x) = \begin{cases} 0 & x < X_{(1)} \\ \frac{i-1}{n-1} + \frac{x-X_{(i)}}{(n-1)(X_{(i+1)}-X_{(i)})} & X_{(i)} \leq x < X_{(i+1)} \\ 1 & X_{(n)} \leq x \end{cases}$$

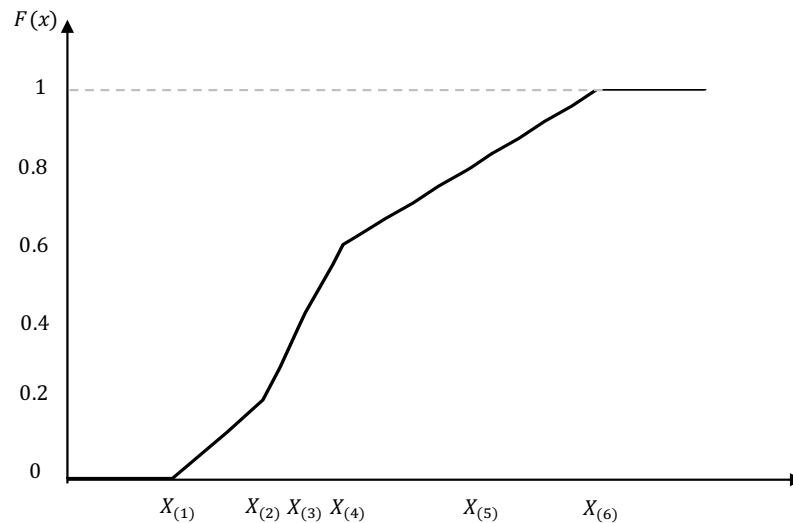
# Empirical distributions

- An empirical distribution is specified directly by the observed data

➤ Continuous random variables

- Case I: The individual original observations  $X_1, X_2, \dots, X_n$  are known

Illustration for  $n = 6$  with  $X_1 = 20, X_2 = 37, X_3 = 44, X_4 = 52, X_5 = 76, X_6 = 100$



Note that  $F(x)$  rises most rapidly over those ranges of  $x$  in which the  $X_j$ 's are most densely distributed

Disadvantages:

- Not possible to generate random values outside range of observed values
- The mean of  $F(x)$  is not equal to the sample mean  $\bar{X}(n)$  of the  $X_j$ 's

# Empirical distributions

- An empirical distribution is specified directly by the observed data

➤ Continuous random variables (not in your slides)

- Case I: The individual original observations  $X_1, X_2, \dots, X_n$  are known

Illustration for  $n = 6$  with  $X_1 = 89, X_2 = 76, X_3 = 15, X_4 = 34, X_5 = 37, X_6 = 29$

$$i = 1 \quad X_{(1)} = 15 \quad F(x = 15) = \frac{1-1}{6-1} + \frac{15-X_{(1)}}{(6-1)(X_{(1+1)}-X_{(1)})} = \frac{1-1}{6-1} = 0$$

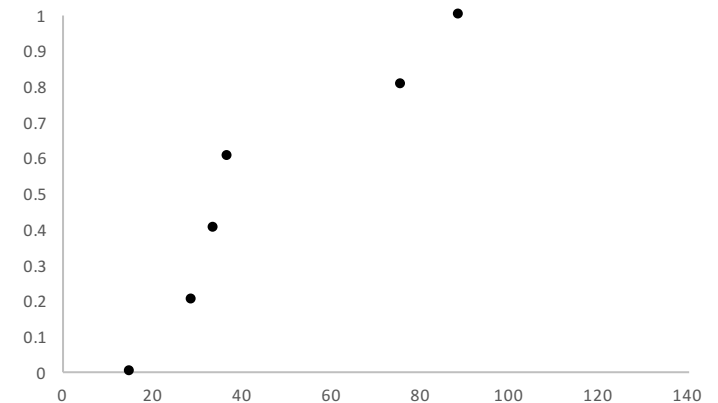
$$i = 2 \quad X_{(2)} = 29 \quad F(x = 29) = \frac{2-1}{6-1} = 0.2$$

$$i = 3 \quad X_{(3)} = 34 \quad F(x = 34) = \frac{3-1}{6-1} = 0.4$$

$$i = 4 \quad X_{(4)} = 37 \quad F(x = 37) = \frac{4-1}{6-1} = 0.6$$

$$i = 5 \quad X_{(5)} = 76 \quad F(x = 76) = \frac{5-1}{6-1} = 0.8$$

$$i = 6 \quad X_{(6)} = 89 \quad F(x = 89) = \frac{6-1}{6-1} = 1$$



# Empirical distributions

- An empirical distribution is specified directly by the observed data

➤ Continuous random variables (not in your slides)

- Case I: The individual original observations  $X_1, X_2, \dots, X_n$  are known

Illustration for  $n = 6$  with  $X_1 = 89, X_2 = 76, X_3 = 15, X_4 = 34, X_5 = 37, X_6 = 29$

$$i = 1 \quad X_{(1)} = 15 \quad F(x = 15) = \frac{1-1}{6-1} + \frac{15-X_{(1)}}{(6-1)(X_{(1+1)}-X_{(1)})} = \frac{1-1}{6-1} = 0$$

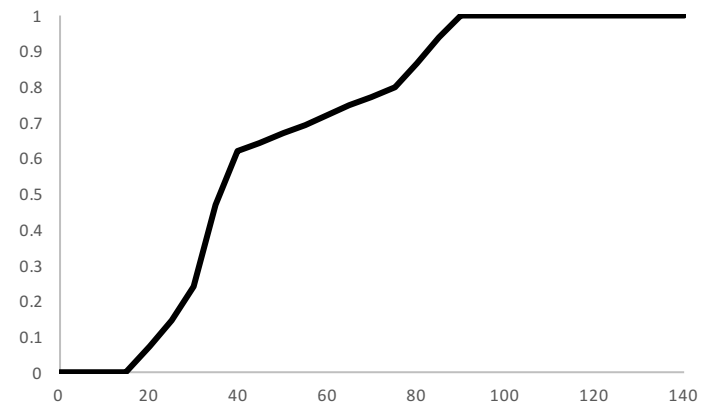
$$i = 2 \quad X_{(2)} = 29 \quad F(x = 15) = \frac{2-1}{6-1} = 0.2$$

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$$i = 4 \quad X_{(4)} = 37 \quad F(x = 15) = \frac{4-1}{6-1} = 0.6$$

$$i = 5 \quad X_{(5)} = 76 \quad F(x = 15) = \frac{5-1}{6-1} = 0.8$$

$$i = 6 \quad X_{(6)} = 89 \quad F(x = 15) = \frac{6-1}{6-1} = 1$$



# Empirical distributions

---

- An empirical distribution is specified directly by the observed data

➤ Continuous random variables (not in your slides)

- Case I: The individual original observations  $X_1, X_2, \dots, X_n$  are known

Illustration for  $n = 6$  with  $X_1 = 89, X_2 = 76, X_3 = 15, X_4 = 34, X_5 = 37, X_6 = 29$

Some example probabilities:

$$F(x = 5) = 0$$

as  $x = 5 < X_{(1)}$

$$F(x = 36) = \frac{3-1}{6-1} + \frac{36-X_{(3)}}{(6-1)(X_{(4)}-X_{(3)})} = 0.4 + \frac{36-34}{(6-1)(37-34)} = 0.533$$

as  $X_{(3)} \leq x = 36 < X_{(4)}$

$$F(x = 94) = 1$$

as  $x = 94 \geq X_{(6)}$

$$F(x = 87) = \frac{5-1}{6-1} + \frac{87-X_{(5)}}{(6-1)(X_{(6)}-X_{(5)})} = 0.8 + \frac{87-76}{(6-1)(89-76)} = 0.969$$

as  $X_{(5)} \leq x = 87 < X_{(6)}$

# Empirical distributions

---

- An empirical distribution is specified directly by the observed data

➤ Continuous random variables

- Case 2: The grouped data or the histogram is known, the values of individual  $X_j$ 's are not known

The histogram reveals the number of observations that fall in each of the specified intervals.

Suppose that the  $n$  observations are grouped into  $k$  adjacent intervals, i.e.  $[a_0, a_1), [a_1, a_2), \dots, [a_{k-1}, a_k)$  so that the  $j$ th interval contains  $n_j$  observations where  $n_1 + n_2 + \dots + n_k = n$

Define the cumulative density distribution  $G(x)$

$$G(x) = \begin{cases} 0 & x < a_0 \\ G(a_{j-1}) + \frac{x-a_{j-1}}{(a_j-a_{j-1})} [G(a_j) - G(a_{j-1})] & a_{j-1} \leq x < a_j \\ 1 & a_k \leq x \end{cases}$$

$$\text{with } G(a_0) = 0 \text{ and } G(a_j) = \frac{n_1 + n_2 + \dots + n_j}{n}$$

# Empirical distributions

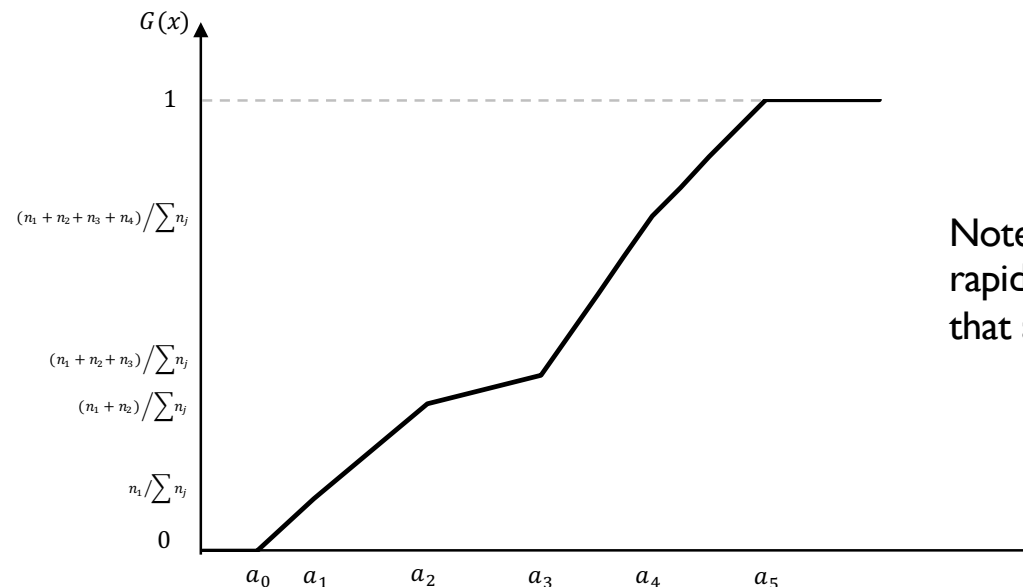
- An empirical distribution is specified directly by the observed data

➤ Continuous random variables

- Case 2: The grouped data or the histogram is known, the values of individual  $X_j$ 's are not known

Example

| $[a_{j-1}, a_j)$ | $n_j$ |
|------------------|-------|
| [10,20)          | 15    |
| [20,40)          | 27    |
| [40,60)          | 8     |
| [60,80)          | 45    |
| [80,100)         | 33    |



Note that  $G(x)$  rises most rapidly over those ranges of  $x$  that are most dense

Disadvantages:

- Not possible to generate random values outside range of observed values



# Empirical distributions

- An empirical distribution is specified directly by the observed data

➤ Continuous random variables

- Case 2: The grouped data or the histogram is known, the values of individual  $X_j$ 's are not known

Example

(not in your slides)

| j | $a_{j-1}$ | $a_j$ | $n_j$ |
|---|-----------|-------|-------|
| 1 | 10        | 25    | 8     |
| 2 | 25        | 30    | 12    |
| 3 | 30        | 60    | 10    |
| 4 | 60        | 80    | 10    |

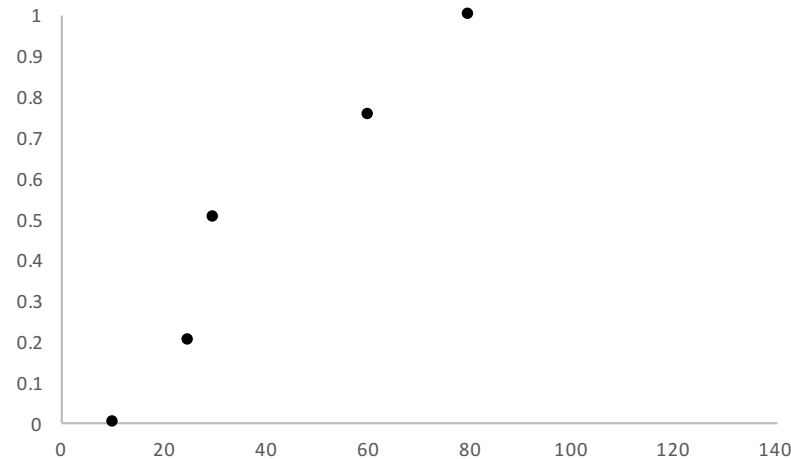
$$a_0: G(x = 10) = 0$$

$$a_1: G(x = 25) = \frac{8}{40} = 0.2$$

$$a_2: G(x = 30) = \frac{8+12}{40} = 0.5$$

$$a_2: G(x = 60) = \frac{8+12+10}{40} = 0.75$$

$$a_4: G(x = 80) = \frac{8+12+10+10}{40} = 1$$



# Empirical distributions

- An empirical distribution is specified directly by the observed data

➤ Continuous random variables

- Case 2: The grouped data or the histogram is known, the values of individual  $X_j$ 's are not known

Example

(not in your slides)

| j | $a_{j-1}$ | $a_j$ | $n_j$ |
|---|-----------|-------|-------|
| 1 | 10        | 25    | 8     |
| 2 | 25        | 30    | 12    |
| 3 | 30        | 60    | 10    |
| 4 | 60        | 80    | 10    |

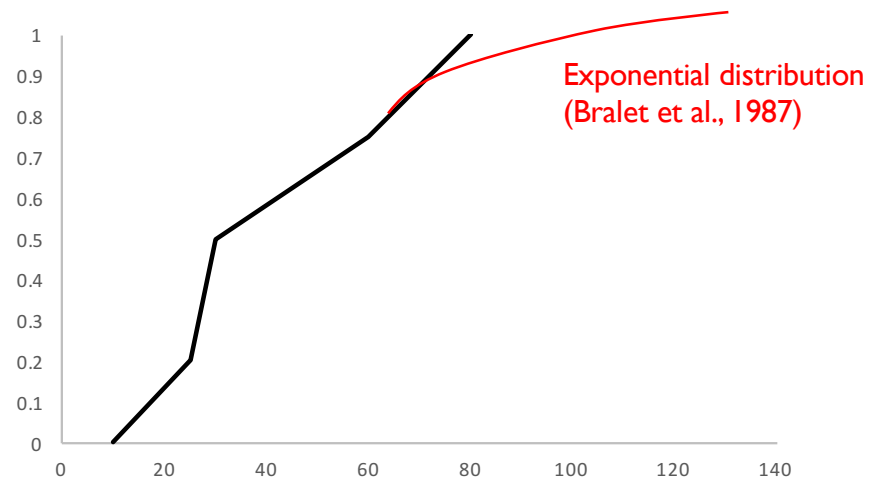
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$$a_2: G(x = 30) = \frac{8+12}{40} = 0.5$$

$$a_2: G(x = 60) = \frac{8+12+10}{40} = 0.75$$

$$a_4: G(x = 80) = \frac{8+12+10+10}{40} = 1$$



# Empirical distributions

---

- An empirical distribution is specified directly by the observed data

- Continuous random variables

- Case 2: The grouped data or the histogram is known, the values of individual  $X_j$ 's are not known

Some example probabilities

*(not in your slides)*

$$G(x = 5) = 0$$

$$\text{as } x = 5 < a_0$$

$$G(x = 15) = G(a_0) + \frac{15-a_0}{(a_1-a_0)} [G(a_1) - G(a_0)] = 0 + \frac{15-10}{(25-10)} [0.2 - 0] = 0.067$$

$$\text{as } a_0 = 10 \leq x = 15 < a_1 = 25$$

$$G(x = 56) = G(a_2) + \frac{56-a_2}{(a_3-a_2)} [G(a_3) - G(a_2)] = 0.5 + \frac{56-30}{(60-30)} [0.75 - 0.5] = 0.717$$

$$\text{as } a_2 = 30 \leq x = 56 < a_3 = 60$$

| j | $a_{j-1}$ | $a_j$ | $\eta_j$ |
|---|-----------|-------|----------|
| 1 | 10        | 25    | 8        |
| 2 | 25        | 30    | 12       |
| 3 | 30        | 60    | 10       |
| 4 | 60        | 80    | 10       |

# Empirical distributions

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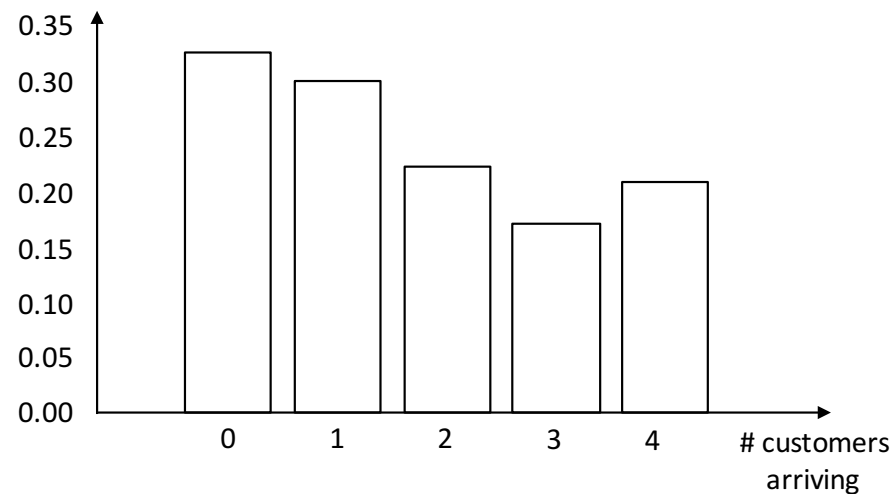
- An empirical distribution is specified directly by the observed data

- Discrete random variables

- Case I: The individual original observations  $X_1, X_2, \dots, X_n$  are known

For each possible value  $x$ , an empirical mass function  $p(x)$  can be defined to be the proportion of the  $X_j$ 's that are equal to  $x$ .

Example: # customers arriving in supermarket per five-minute interval



# Empirical distributions

---

- An empirical distribution is specified directly by the observed data

➤ Discrete random variables

- Case I: The individual original observations  $X_1, X_2, \dots, X_n$  are known (not in your slides)

| i  | $X_i$ |
|----|-------|
| 1  | 0     |
| 2  | 2     |
| 3  | 1     |
| 4  | 0     |
| 5  | 0     |
| 6  | 1     |
| 7  | 2     |
| 8  | 2     |
| 9  | 2     |
| 10 | 2     |

$$p(X = x_i) = \frac{\#observations=x_i}{total \# observations}$$

$$F(X = x_i) = \sum_{x \leq x_i} p(X = x)$$

$$p(X = 0) = \frac{3}{10}$$

$$F(X = 0) = \frac{3}{10}$$

$$p(X = 1) = \frac{2}{10}$$

$$F(X = 1) = \frac{5}{10}$$

$$p(X = 2) = \frac{5}{10}$$

$$F(X = 2) = 1$$

# Empirical distributions

- An empirical distribution is specified directly by the observed data

➤ Discrete random variables

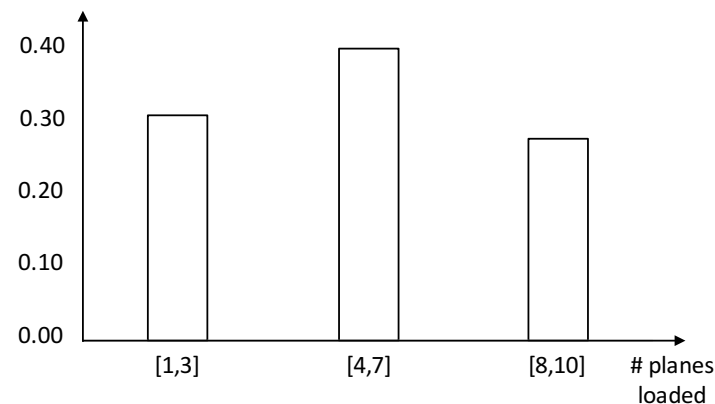
- Case 2: The grouped data or the histogram is known, the values of individual  $X_j$ 's are not known

Step 1: Define a mass function such that the sum of the  $p(x)$ 's over all possible values of  $x$  in an interval is equal to the proportion of the  $X_j$ 's in that interval.

Step 2: How the individual  $p(x)$ 's are allocated for the possible values of  $x$  within an interval is essentially arbitrary.

Example: Daily workload of an airplane loading company (low – medium – high)

| Workload | $[a_{j-1}, a_j]$ | $n_j$ |
|----------|------------------|-------|
| Low      | [1,3]            | 10    |
| Medium   | [4,7]            | 13    |
| High     | [8,10]           | 9     |



Note that it is not clear how the individual  $p(x)$  are allocated for the different values for  $x$ .

# Empirical distributions

- An empirical distribution is specified directly by the observed data

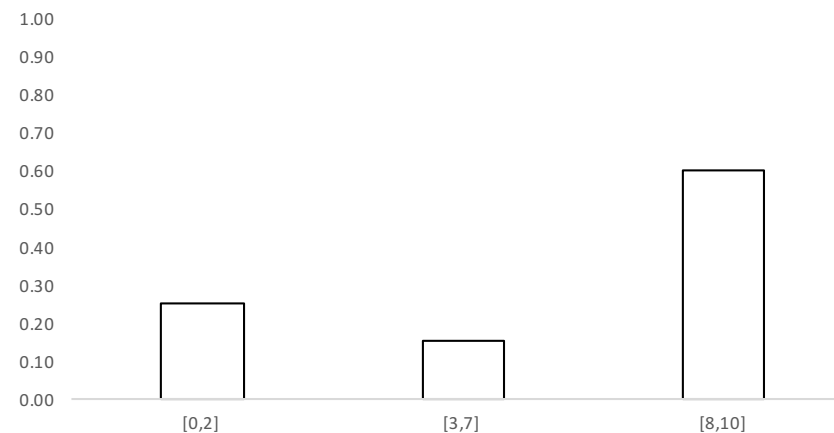
➤ Discrete random variables

- Case 2: The grouped data or the histogram is known, the values of individual  $X_j$ 's are not known

Example

(not in your slides)

| $[a_{j-1}, a_j]$ | $n_j$ | $p_j$ | $F_j$ |
|------------------|-------|-------|-------|
| [0,2]            | 5     | 0.25  | 0.25  |
| [3,7]            | 3     | 0.15  | 0.40  |
| [8,10]           | 12    | 0.60  | 1     |



Note that it is not clear how the individual  $p(x)$  are allocated for the different values for  $x$ .

# Parametrisation of probability distributions

---

- Relevant parameters for continuous distributions
  - A location parameter  $\gamma$  specifies an abscissa (x axis) location point of a distribution's range of values
    - $\gamma$  is usually the midpoint or lower endpoint of the distribution's range
    - A change in  $\gamma$  shifts the distribution to the left or right without otherwise changing
  - A scale parameter  $\beta$  determines the scale of measurement of the values in the range of distribution
    - A change in  $\beta$  compresses or expands the distribution without altering its basic form
    - E.g. The standard deviation  $\sigma$  is a scale parameter for the normal distribution
  - A shape parameter  $\alpha$  determines the basic form of a distribution
    - A change in  $\alpha$  alters the distribution's properties fundamentally (e.g. Skewness)
    - Exponential and normal distributions do not have a shape parameter; beta distribution has two shape parameters



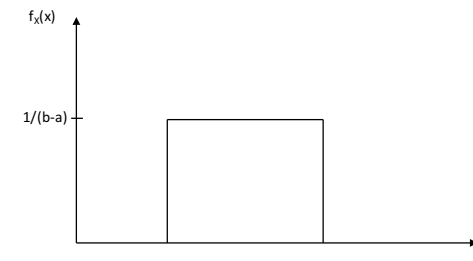
# Parametrisation of probability distributions

- Some examples

- The Uniform distribution

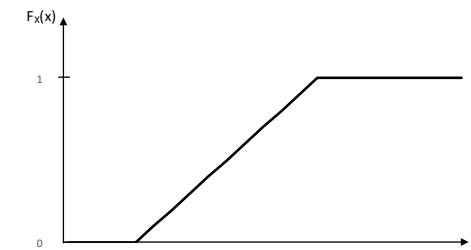
- $X$  is a continuous random variable within the range  $[a,b]$
- Density function

$$f_X(x) = \begin{cases} c & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases} \quad \text{and } c = \frac{1}{b-a}$$



- Cumulative distribution function

$$F_X(x) = \int_{-\infty}^x f_X(t) dt = \frac{1}{b-a} t \Big|_{-\infty}^x = \begin{cases} 0 & x \leq a \\ (x-a)/(b-a) & a \leq x \leq b \\ 1 & b \leq x \end{cases}$$



- Parametrisation

$a$  is location parameter  $\gamma$

$b-a$  is scale parameter  $\beta$

# Parametrisation of probability distributions

- Some examples

- The Exponential distribution

- $X$  is a continuous random variable with  $\lambda > 0$
- Density function

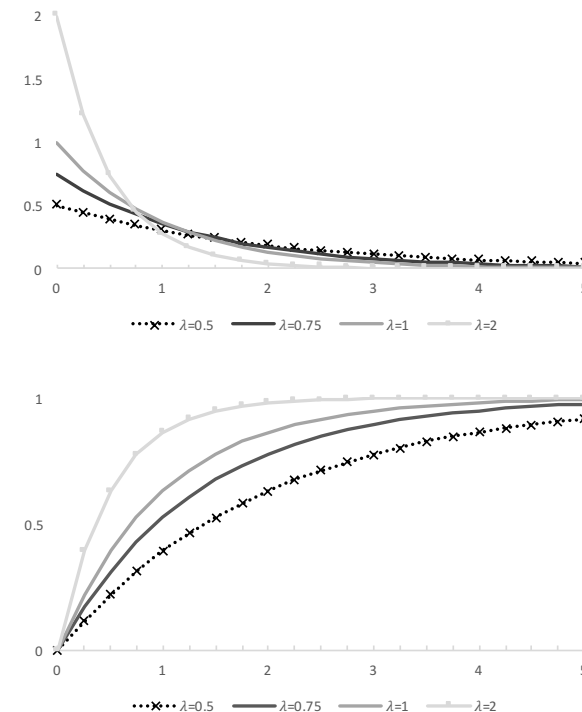
$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- Cumulative distribution function

$$F(x) = \begin{cases} 1 - e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- Parametrisation

$\frac{1}{\lambda}$  is scale parameter  $\beta$



# Parametrisation of probability distributions

- Some examples

- The Gamma distribution

- $X$  is a continuous random variable
- The gamma function

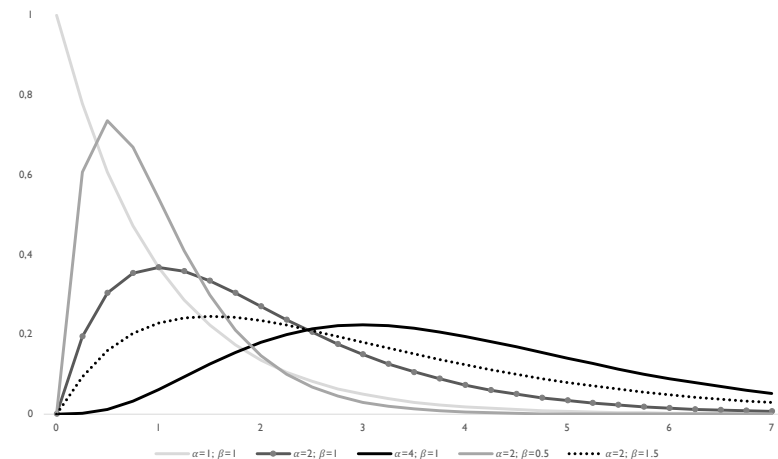
$$\Gamma(\alpha) = \int_0^{\infty} y^{\alpha-1} e^{-y} dy$$

- Probability density function

$$f_X(x) = f(x; \beta, \alpha) = \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta} \quad \text{for } x > 0$$

with  $\alpha$  the shape parameter

$\beta$  the scale parameter



# Theoretical distributions

---

- Specifying theoretical distributions

- Relevant steps

Step 1: Hypothesizing families of distributions

Model uncertainty

Step 2: Estimation of parameters

Parameter uncertainty

=> Input-model  
uncertainty

Step 3: Determining how representative the fitted distributions are

- Use of specialised software (Expertfit, Matlab, R, ...)

- Automated fitting
    - Fitting is done with a large number of distributions and the best fit is proposed
    - If the best theoretical distribution gives a bad fit, the empirical distribution is proposed

# Theoretical distributions

---

- Step 1: Hypothesizing families of distributions
  - Use of prior knowledge to select a modelling distribution
    - Examples
      - Service times cannot be normally distributed since a random value from any normal distribution can be negative
      - The proportion of defective items in a large batch should not be assumed to have a gamma distribution since proportions should be between 0 and 1 whereas gamma distributions have no upper bound
    - Further confirmation is recommended
      - Step 1: Use different heuristics or guidelines to choose appropriate families of distributions
      - Step 2: Estimate parameters via maximum-likelihood estimators
      - Step 3: Perform heuristics and goodness-of-fit hypothesis tests to determine the quality of the fitted distributions

# Theoretical distributions

- Step I: Hypothesizing families of distributions

- Summary statistics

| Function                                                            | Sample estimate                                                                                                                                         | Continuous (C)<br>or Discrete (D) | Comments                                              |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-------------------------------------------------------|
| Minimum, maximum                                                    | $X_{(1)}, X_{(n)}$                                                                                                                                      | C,D                               | $[X_{(1)}, X_{(n)}]$ is a rough estimate of the range |
| Mean $\mu$                                                          | $\bar{X}(n)$                                                                                                                                            | C,D                               | Measure of central tendency                           |
| Median $x_{0.5}$                                                    | $\hat{x}_{0.5} = \begin{cases} X_{((n+1)/2)} & \text{if } n \text{ is odd} \\ [X_{(n/2)} + X_{((n/2)+1)}]/2 & \text{if } n \text{ is even} \end{cases}$ | C,D                               | Alternative measure of central tendency               |
| Variance $\sigma^2$                                                 | $S^2(n)$                                                                                                                                                | C,D                               | Measure of variability                                |
| Coefficient of variation, $cv = \frac{\sqrt{\sigma^2}}{\bar{X}(n)}$ | $\widehat{cv}(n) = \frac{\sqrt{S^2}}{\bar{X}(n)}$                                                                                                       | C                                 | Alternative measure of variability                    |
| Lexis ratio $\tau = \frac{\sigma^2}{\mu}$                           | $\hat{\tau}(n) = \frac{S^2(n)}{\bar{X}(n)}$                                                                                                             | D                                 | Alternative measure of variability                    |
| Skewness, $v = \frac{E[(X - \mu)^3]}{(\sigma^2)^{3/2}}$             | $\hat{v}(n) = \frac{n^2}{(n-1)(n-2)} \frac{\sum_{i=1}^n [X_i - \bar{X}(n)]^3 / n}{(S^2(n))^{3/2}}$                                                      | C,D                               | Measure of symmetry                                   |

# Theoretical distributions

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- Step 1: Hypothesizing families of distributions

- Summary statistics

- These functions might suggest an appropriate distribution family

- For a symmetric distribution (e.g. normal distribution) the mean is equal to the median, i.e. when the estimates  $\bar{X}(n)$  and  $\hat{x}_{0.5}$  are almost equal

- The coefficient of variation  $cv$  provides information about the form of a continuous distribution

$cv = 1$  for an exponential distribution regardless of the scale parameter  $\beta$

$cv > 1, = 1, < 1$  for the gamma and Weibull distributions when the shape parameter  $\alpha < 1, = 1, > 1$

- The lexis ratio  $\tau$  is similar to the coefficient of variation for discrete distributions

$\tau = 1$  Poisson distribution

$\tau < 1$  Binomial distribution

$\tau > 1$  Negative binomial distribution

- Skewness  $v$  is a measure of the symmetry of a distribution

$v = 0$  for symmetric distributions (e.g. normal distribution)

$v > 0$  the distribution is skewed to the right

$v < 0$  the distribution is skewed to the left

# Theoretical distributions

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- Step I: Hypothesizing families of distributions

- Histograms

- What? A graphical estimate of the plot of the density function corresponding to the data  $X_1, X_2, \dots, X_n$
- Continuous distributions
  - Construct  $k$  disjoint adjacent intervals, i.e.  $[a_0, a_1), [a_1, a_2), \dots, [a_{k-1}, a_k)$  with the same width  $\Delta a = a_j - a_{j-1}$
  - Let  $h_j$  be the proportion of the  $X_j$ 's that are in the  $j$ th interval  $[a_{j-1}, a_j)$

$$h(x) = \begin{cases} 0 & x < a_0 \\ h_j & a_{j-1} \leq x < a_j \\ 0 & a_k \leq x \end{cases} \quad \text{for } j = 1, 2, \dots, k$$

- Discrete distributions
  - Let  $h_j$  be the proportion of the  $X_j$ 's that are equal to  $x_j$
  - $h_j$  is an unbiased estimator of  $p(x_j)$
- The plot of  $h(x)$  is compared with densities of various distributions on the basis of the shape



# Theoretical distributions

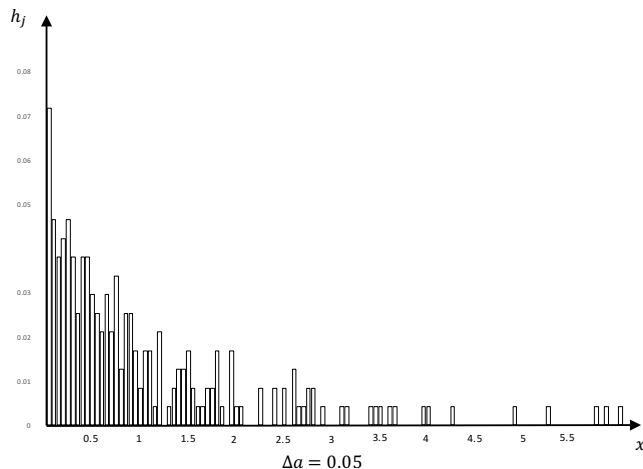
- Step I: Hypothesizing families of distributions

- Histograms

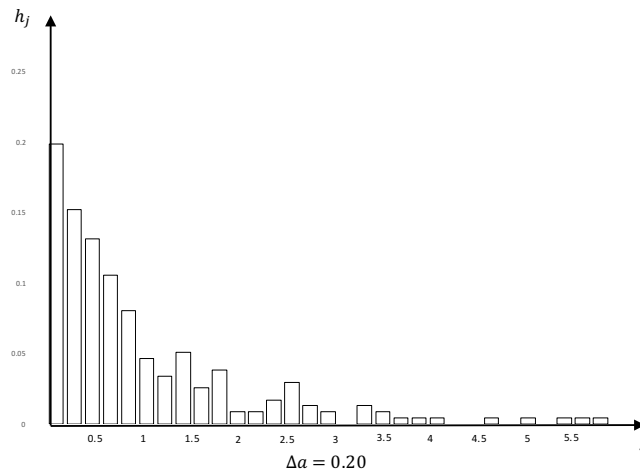
- How to select the number of intervals  $k$  (or equivalently their width  $\Delta a$ )?

Continuous distributions: Try several different values  $\Delta a$  and select the smallest one that gives a smooth histogram

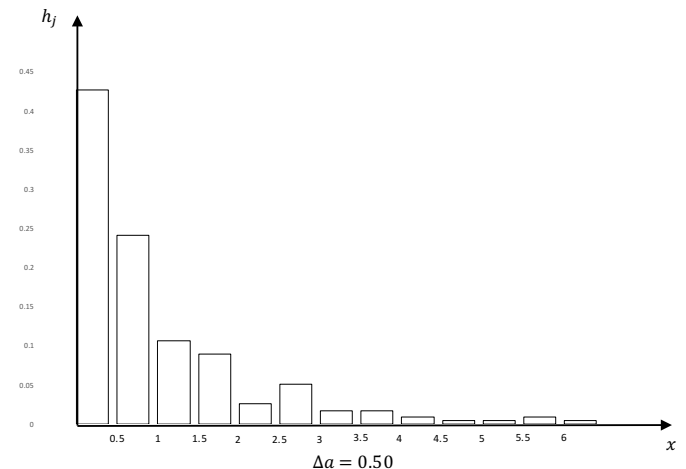
- Example: 237 random observations of an exponential distribution ( $\lambda = 1$ )



$\Delta a$  too small  
“ragged” shape



“smooth” shape



$\Delta a$  too large  
“block-like” shape

# Theoretical distributions

- Step I: Hypothesizing families of distributions

- Quantile summaries and boxplots

- Useful to determine the symmetry or skewness of a distribution

$F(x)$  is the distribution function for a continuous variable

$F(x)$  is (strictly) increasing when  $0 < F(x) < 1$ , i.e. if  $x_1 < x_2$  then  $0 < F(x_1) \leq F(x_2) < 1$

- q-quantile function:  $F(x_q) = q$  for  $0 < q < 1$  or  $F^{-1}(q) = x_q$  (with  $F^{-1}(x)$  is the inverse function of  $F(x)$ )
- Structure of the quantile summary for sample  $X_1, X_2, \dots, X_n$

| Quantile  |                         | Depth             | Sample values           | Midpoint                    |
|-----------|-------------------------|-------------------|-------------------------|-----------------------------|
| Median    | $x_{0.5}$               | $i = (n + 1)/2$   | $X_{(i)}$               | $X_{(i)}$                   |
| Quartiles | $x_{0.25}$ $x_{0.75}$   | $j = ([i] + 1)/2$ | $X_{(j)}$ $X_{(n-j+1)}$ | $[X_{(j)} + X_{(n-j+1)}]/2$ |
| Octiles   | $x_{0.125}$ $x_{0.875}$ | $k = ([j] + 1)/2$ | $X_{(k)}$ $X_{(n-k+1)}$ | $[X_{(k)} + X_{(n-k+1)}]/2$ |
| Extremes  | $x_{0.001}$ $x_{0.999}$ | 1                 | $X_{(1)}$ $X_{(n)}$     | $[X_{(1)} + X_{(n)}]/2$     |

Symmetric – midpoints are equal  
 Skewed to the left – midpoints are decreasing  
 Skewed to the right – midpoints are increasing

- Boxplot: Graphical representation of a quantile summary (50% of the observations fall within the box)

# Theoretical distributions

- Step I: Hypothesizing families of distributions

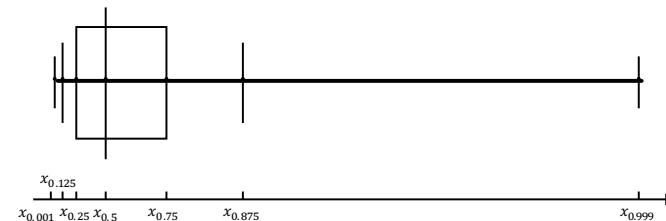
- Quantile summaries and boxplots

- Example: 237 random observations of an exponential distribution ( $\lambda = 1$ )

| Function                 | Value  |
|--------------------------|--------|
| Minimum                  | 0.0014 |
| Maximum                  | 7.1382 |
| Mean                     | 1.0286 |
| Median                   | 0.6292 |
| Variance                 | 1.3313 |
| Coefficient of variation | 1.1218 |
| Skewness                 | 2.1912 |

| Quantile  | Depth | Sample values |        | Midpoint |
|-----------|-------|---------------|--------|----------|
| Median    | 119   | 0.6291        |        | 0.6291   |
| Quartiles | 60    | 0.2710        | 1.3716 | 0.8213   |
| Octiles   | 30.5  | 0.1054        | 2.2979 | 1.2017   |
| Extremes  | 1     | 0.0014        | 7.1381 | 3.5698   |

Boxplot



# Theoretical distributions

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- Step 2: Estimation of parameters

- Estimation of distribution parameters is based on  $X_1, X_2, \dots, X_n$

- Estimator: Maximum-likelihood estimators (MLEs)

- Consider a discrete (or continuous) distribution for the data with one unknown parameter
    - Suppose  $p_\theta(x)$  denotes the probability mass function
    - The likelihood function:  $L(\theta) = p_\theta(X_1)p_\theta(X_2) \dots p_\theta(X_n)$

with  $L(\theta)$  the probability (likelihood) of obtaining our observed data if  $\theta$  is the value of the unknown parameter

$\hat{\theta}$  the optimal value of  $\theta$  that maximizes  $L(\theta)$ , i.e.  $L(\hat{\theta}) \geq L(\theta), \forall \theta$

# Theoretical distributions

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- Step 2: Estimation of parameters

- Estimator: Maximum-likelihood estimators (MLEs)

- Example: Hypothesised distribution = Normal distribution ( $f_X(x) = N(\mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$ )

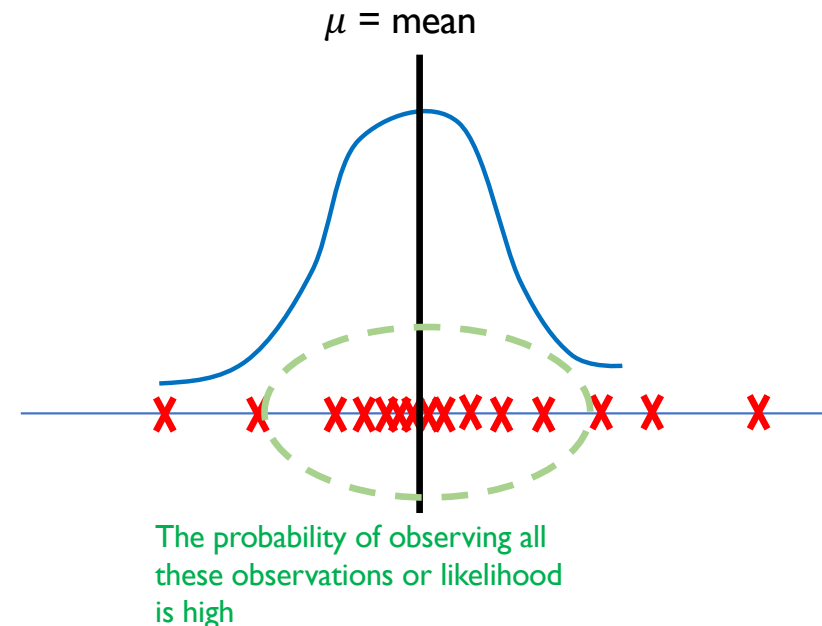
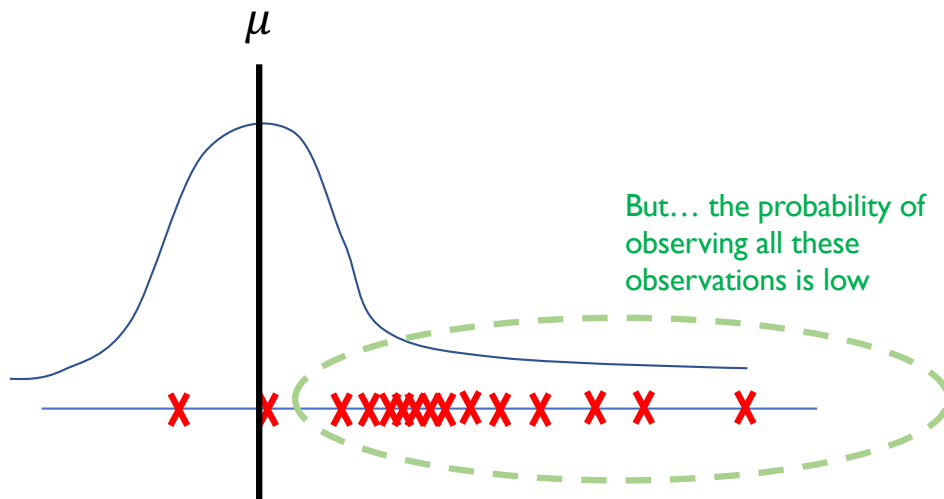


# Theoretical distributions

- Step 2: Estimation of parameters

- Estimator: Maximum-likelihood estimators (MLEs)

- Example: Hypothesised distribution = Normal distribution  $(f_X(x) = N(\mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2})$

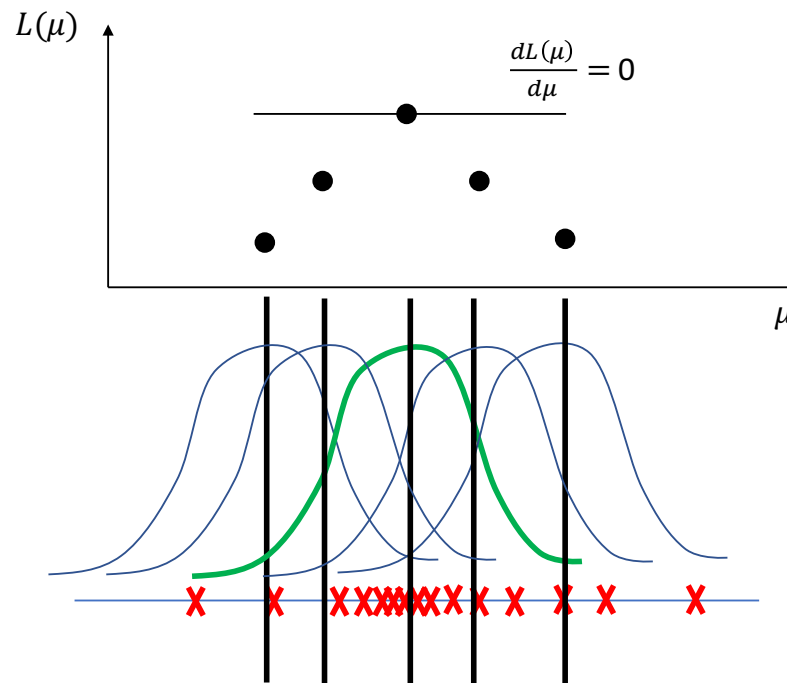


# Theoretical distributions

- Step 2: Estimation of parameters

- Estimator: Maximum-likelihood estimators (MLEs)

- Example: Hypothesised distribution = Normal distribution ( $f_X(x) = N(\mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$ )



# Theoretical distributions

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- Step 2: Estimation of parameters

- Estimator: Maximum-likelihood estimators (MLEs)

- Example: Hypothesised distribution = Bernoulli distribution *(not in your slides)*

- Given samples: 0 – 1 – 0 – 0 – 1 – 0 (6 observations)
- Distribution: Bernoulli distribution with form  $P(X = 0) = 1 - \mu$  and  $P(X = 1) = \mu$
- What is the maximum likelihood estimate of  $\mu$ ?

The likelihood function

$$\begin{aligned}L(\mu) &= P(X = 0)P(X = 1)P(X = 0)P(X = 0)P(X = 1)P(X = 0) \\&= (1 - \mu) \mu (1 - \mu) (1 - \mu) \mu (1 - \mu) \\&= (1 - \mu)^4 \mu^2\end{aligned}$$

$$\Rightarrow \max_{\mu} L(\mu) = (1 - \mu)^4 \mu^2$$

$$\text{s.t. } \mu > 0$$

$$\Rightarrow \text{The log-likelihood function } \log L(\mu) = \log (1 - \mu)^4 \mu^2 = 4 \log(1 - \mu) + 2 \log \mu$$

(since the logarithm is strictly increasing, maximising  $L(\lambda)$  is equivalent to maximising  $\log L(\mu)$ )

$$\Rightarrow \frac{d \log L(\mu)}{d\mu} = \frac{4}{1-\mu} (-1) + \frac{2}{\mu} = 0 \quad (\text{the first derivative is set to 0; note that the second derivative } \frac{d^2 \log L(\mu)}{d\mu^2} \leq 0 \text{ since } \mu > 0)$$

$$\Rightarrow \mu = \frac{1}{3}$$



# Theoretical distributions

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- Step 2: Estimation of parameters

- Estimator: Maximum-likelihood estimators (MLEs)

- Example: 237 random observations of an exponential distribution ( $\lambda = 1$ )

- For the exponential distribution  $\theta = \lambda$  ( $\lambda > 0$ ) and  $f_{\lambda}(x) = \lambda e^{-\lambda x}$  for  $x \geq 0$

- The likelihood function
$$L(\lambda) = (\lambda e^{-\lambda X_1})(\lambda e^{-\lambda X_2}) \dots (\lambda e^{-\lambda X_n})$$
$$= (\lambda^n e^{-\lambda \sum_{i=1}^n X_i})$$

$$\Rightarrow \max_{\lambda} L(\lambda) = (\lambda^n e^{-\lambda \sum_{i=1}^n X_i})$$

$$\text{s.t.} \quad \lambda > 0$$

$$\Rightarrow \text{The log-likelihood function} \quad l(\lambda) = \ln L(\lambda) = \ln(\lambda^n e^{-\lambda \sum_{i=1}^n X_i}) = n \ln(\lambda) - \lambda \sum_{i=1}^n X_i$$

(since the logarithm is strictly increasing, maximising  $L(\lambda)$  is equivalent to maximising  $l(\lambda)$ )

$$\Rightarrow \frac{dl}{d\lambda} = \frac{n}{\lambda} - \sum_{i=1}^n X_i = 0 \quad (\text{the first derivative is set to 0; note that the second derivative } \frac{d^2l}{d\lambda^2} \leq 0 \text{ since } \lambda > 0)$$

$$\Rightarrow \lambda = \frac{n}{\sum_{i=1}^n X_i} \quad \text{or } 1/\lambda = \frac{\sum_{i=1}^n X_i}{n} \text{ (i.e. the sample mean)}$$

$$\Rightarrow \lambda = 0.9722$$

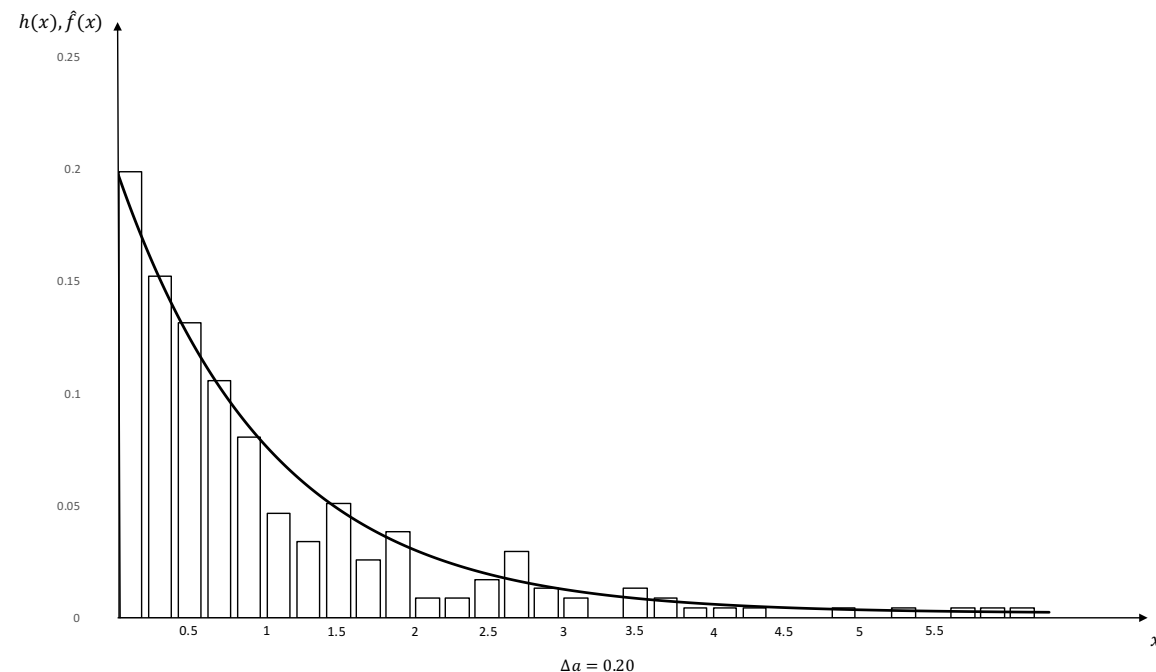
# Theoretical distributions

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- Step 3: Determining how representative the fitted distributions are
  - Find best fit distribution that is accurate enough
  - Techniques for fitting distributions
    - Heuristic or graphical procedures
      - Density-Histogram plots and frequency comparisons
      - Distribution-Function-Differences Plots
      - Probability plots
    - Goodness-of-Fit tests
      - Chi-square test
      - Kolmogorov-Smirnov test

# Theoretical distributions

- Step 3: Determining how representative the fitted distributions are
  - Heuristic or graphical techniques for fitting distributions
    - Density-Histogram plots
      - For continuous data, a density-histogram plot can be made by plotting  $\Delta a \times \hat{f}(x)$  over the histogram  $h(x)$
      - Example: 237 random observations of an exponential distribution ( $\lambda = 1$ )



# Theoretical distributions

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- Step 3: Determining how representative the fitted distributions are

- Heuristic or graphical techniques for fitting distributions

- Frequency comparisons: Graphical comparison of individual probabilities between the histogram of the observed data with the density function of the fitted distribution

- Continuous distribution: Compare  $h_j$  and  $r_j$  in the  $j$ th histogram interval  $j = 1, 2, \dots, k$

Let  $[a_0, a_1), [a_1, a_2), \dots, [a_{k-1}, a_k)$   $k$  disjoint adjacent histogram intervals, each with the same width  $\Delta a = a_j - a_{j-1}$

$h_j$  the observed proportion of the  $X_j$ 's that are in the  $j$ th interval  $[a_{j-1}, a_j)$

$r_j$  the expected proportion of the  $n$  observations that fall in the  $j$ th interval  $[a_{j-1}, a_j)$  according to the fitted distribution, i.e.  $r_j = \int_{a_{j-1}}^{a_j} \hat{f}(x) dx$

- Discrete distribution: Compare  $h_j$  and  $r_j$  for all relevant values of  $x_j$

Let  $x_j$  the relevant values for  $x$

$h_j$  the observed proportion of the  $X_j$ 's that are equal to  $x_j$

$r_j$  the expected proportion of the  $n$  observations that are equal to  $x_j$  according to the fitted distribution, i.e.  $r_j = \hat{p}(x_j)$

=>  $h_j$  and  $r_j$  should closely match for having a good representation for the data by the fitted distribution

# Theoretical distributions

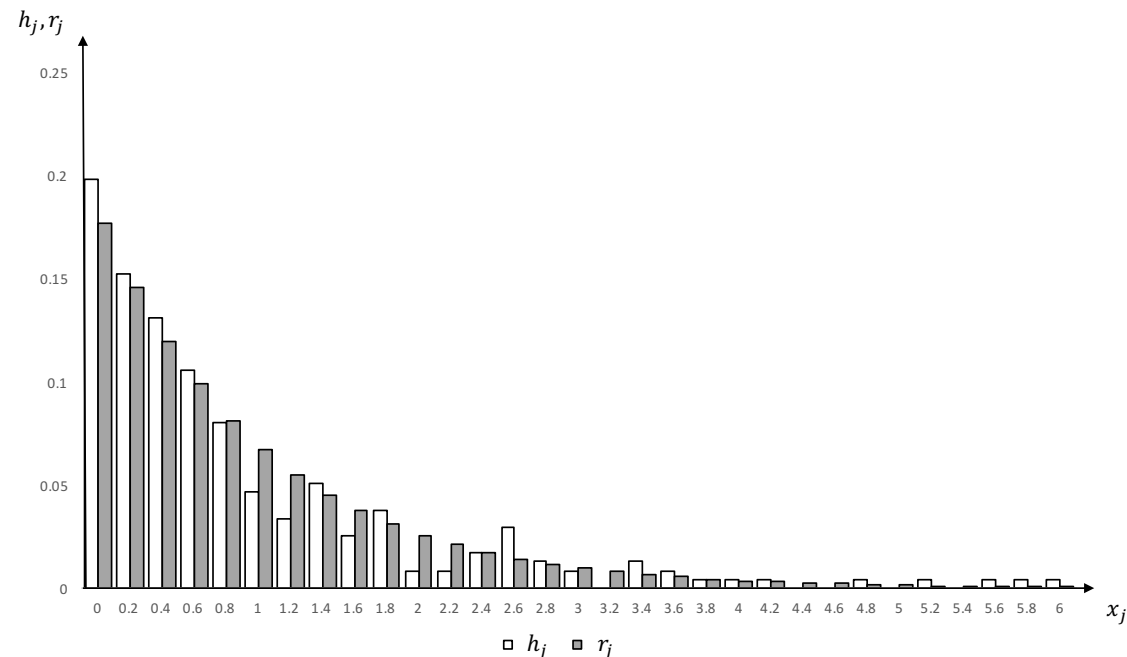
- Step 3: Determining how representative the fitted distributions are

- Heuristic or graphical techniques for fitting distributions

- Frequency comparisons

- Example: 237 random observations of an exponential distribution ( $\lambda = 1$ )

$$r_j = \int_{a_{j-1}}^{a_j} \hat{f}(x) dx \text{ with } \hat{f}(x) = \lambda e^{-\lambda x} \text{ or } r_j = \hat{F}(a_j) - \hat{F}(a_{j-1}) = (1 - e^{-\lambda a_j}) - (1 - e^{-\lambda a_{j-1}}) \quad \text{with } \lambda = 0.9722$$



# Theoretical distributions

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- Step 3: Determining how representative the fitted distributions are

- Heuristic or graphical techniques for fitting distributions

- Distribution-Function-Differences Plots: Graphical comparison of cumulative probabilities between the histogram of the observed data with the density function of the fitted distribution

- Define

Cumulative empirical distribution:  $F_n(x) = \frac{\text{the number of } x_i \text{ values} \leq x}{n}$

Cumulative fitted distribution  $\hat{F}(x)$  is the theoretical probability returning a value  $\leq x$

Distribution-Function-Differences  $\hat{F}(x) - F_n(x)$

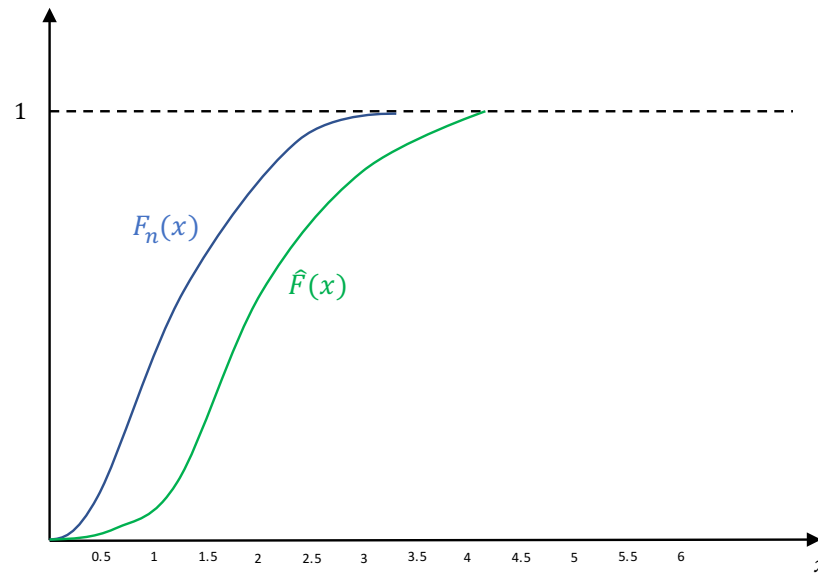
- The smaller the difference, the better the quality of fit

(a perfect fit when this plot is a horizontal line at height 0)

# Theoretical distributions

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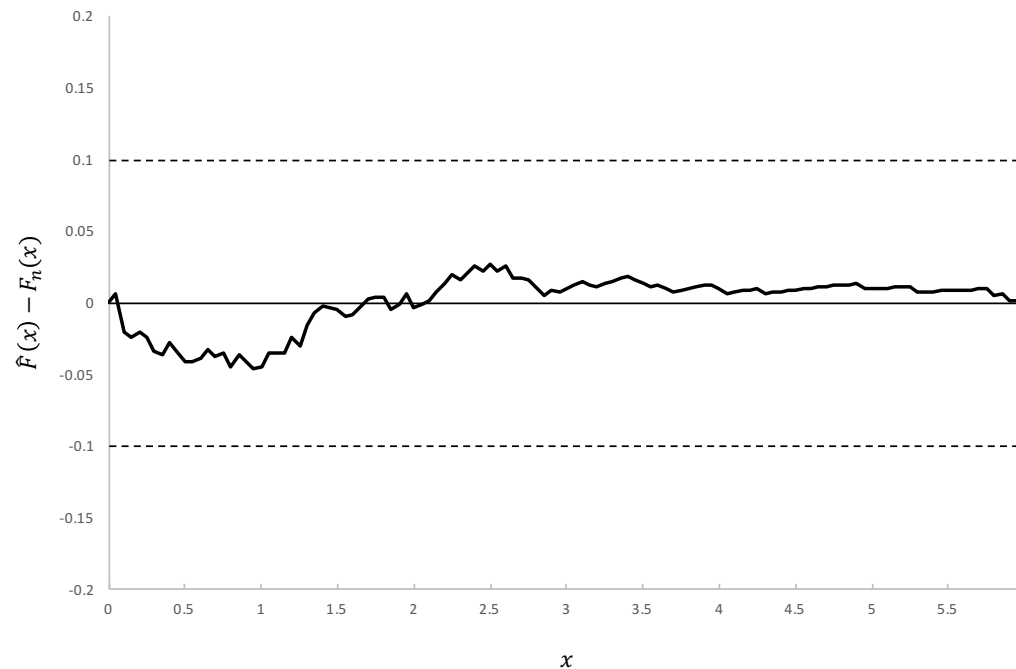
- Step 3: Determining how representative the fitted distributions are
  - Heuristic or graphical techniques for fitting distributions



# Theoretical distributions

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- Step 3: Determining how representative the fitted distributions are
  - Heuristic or graphical techniques for fitting distributions
    - Distribution-Function-Differences Plots
      - Example: 237 random observations of an exponential distribution ( $\lambda = 1$ )





# Theoretical distributions

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- Step 3: Determining how representative the fitted distributions are

- Heuristic or graphical techniques for fitting distributions

- Probability plots: Quantile-Quantile (Q-Q) plot = A graph of the  $q_i$ -quantile of a fitted (model) distribution function  $\hat{F}(X_{(i)})$  versus the empirical distribution

Define

$X_{(i)}$  the  $i$ th smallest of the  $X_j$ 's, i.e.  $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$

$F_n(X_{(i)})$  empirical distribution  $i/n$

$\tilde{F}_n(X_{(i)})$   $F_n(X_{(i)}) - 0.5/n = i - 0.5/n = q_i$ -quantile for  $i = 1, 2, \dots, n$  so that  $0 < q_i < 1$

$\hat{F}(X_{(i)})$  fitted distribution with  $q_i$ -quantile  $= \hat{F}^{-1}(q_i)$

Construct a scatter plot: ( $q_i$ -quantile empirical distribution  $\tilde{F}_n(X_{(i)})$ ,  $q_i$ -quantile fitted distribution  $\hat{F}(X_{(i)})$ )

=> if  $\hat{F}(X_{(i)})$  shows a good fit with  $\tilde{F}_n(X_{(i)})$ , the Q-Q plot is approximately linear with an intercept of 0 and a slope of 1

# Theoretical distributions

- Step 3: Determining how representative the fitted distributions are

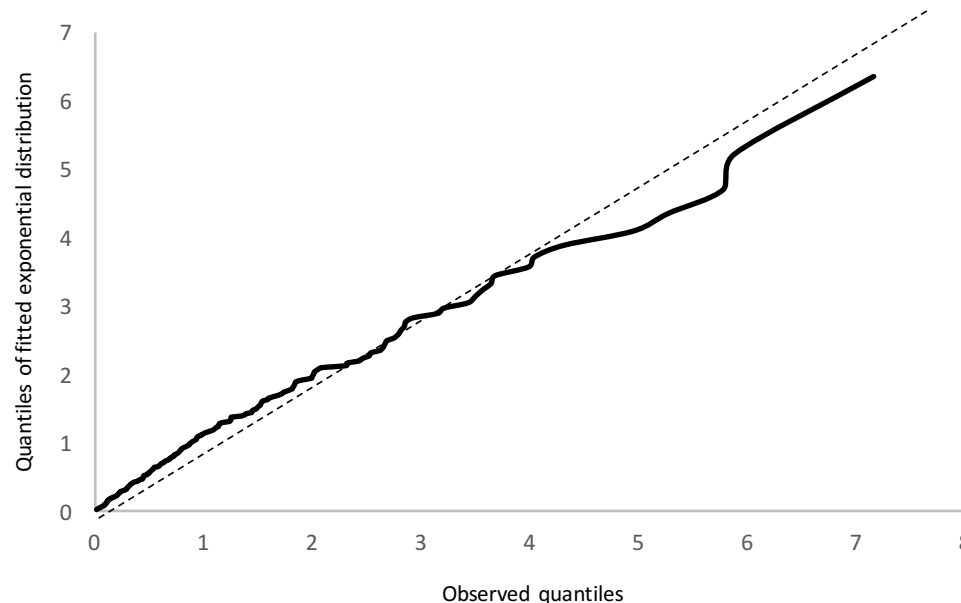
- Heuristic or graphical techniques for fitting distributions

- Probability plots: Quantile-Quantile (Q-Q) plot

- Example: 237 random observations of an exponential distribution ( $\lambda = 1$ )

Calculate  $\tilde{F}_n(X_{(i)}) = i^{-0.5}/n = q_i$

Calculate  $\hat{F}^{-1}(q_i)$       => Draw scatter plot  $(X_{(i)}, \hat{F}^{-1}(q_i))$



Q-Q plot for exponential  
distribution and interarrival  
data

# Theoretical distributions

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- Step 3: Determining how representative the fitted distributions are
  - Heuristic or graphical techniques for fitting distributions
    - Probability plots: Probability-Probability (P-P) plot = A graph of the model probability  $\hat{F}(X_{(i)})$  versus the sample probability  $\tilde{F}_n(X_{(i)}) = q_i$  for  $i = 1, 2, \dots, n$ 
      - => if  $\hat{F}(X_{(i)})$  shows a good fit with  $\tilde{F}_n(X_{(i)})$ , the P-P plot is approximately linear with an intercept of 0 and a slope of 1
      - Construct a scatter plot: (sample probability  $\tilde{F}_n(X_{(i)}) = q_i$ , model probability  $\hat{F}(X_{(i)})$ )
    - Q-Q plot vs P-P plot
      - Q-Q plot amplifies differences between the tails of  $\hat{F}(X_{(i)})$  and  $\tilde{F}_n(X_{(i)})$
      - P-P plot amplifies differences between the middle of  $\hat{F}(X_{(i)})$  and  $\tilde{F}_n(X_{(i)})$

# Theoretical distributions

- Step 3: Determining how representative the fitted distributions are

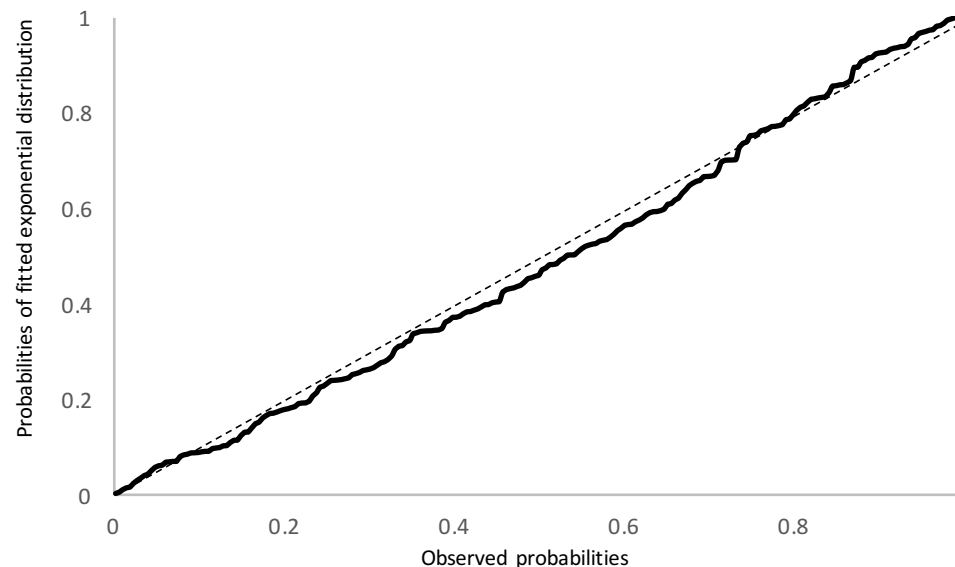
- Heuristic or graphical techniques for fitting distributions

- Probability plots: Probability-Probability (P-P) plot

- Example: 237 random observations of an exponential distribution ( $\lambda = 1$ )

Calculate  $\tilde{F}_n(X_{(i)}) = i^{-0.5}/n = q_i$

Calculate  $\hat{F}(X_{(i)}) = 1 - e^{-\lambda X_{(i)}}$       => Draw scatter plot  $(q_i, \hat{F}(X_{(i)}))$



P-P plot for exponential  
distribution and interarrival  
data

# Theoretical distributions

---

- Step 3: Determining how representative the fitted distributions are

- Goodness-of-Fit tests

- Formally assessment whether  $X_1, X_2, \dots, X_n$  is a independent sample from distribution function  $\hat{F}(x)$

- $H_0$ : The  $X_i$ 's are IID random variables with distribution function  $\hat{F}(x)$

- $H_1$ : The  $X_i$ 's are not IID random variables with distribution function  $\hat{F}(x)$

- Remarks

- For small sample size  $n$ : Techniques are not sensitive to identify (dis)agreements between the data and the fitted distribution
    - For large sample size  $n$ : The null-hypothesis will almost always be rejected

# Theoretical distributions

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- Step 3: Determining how representative the fitted distributions are

- Goodness-of-Fit tests: Chi-square test

- $\chi = \{X_1, \dots, X_n\}$  is a sample drawn from a population with unknown distribution function  $F_X(x)$ 
  - Null Hypothesis ( $H_0$ ):  $F_X(x) = F_0(x)$  for all  $x$  (with  $F_0(x)$  is a completely specified distribution)
  - Alternative hypothesis ( $H_1$ ):  $F_X(x) \neq F_0(x)$  for some  $x$
- $n$  observations grouped into  $k$  mutually exclusive categories (preferably with an equal expected proportion), i.e.  $[a_0, a_1), [a_1, a_2), \dots, [a_{k-1}, a_k)$

with  $n_j$  the number of trial outcomes in  $j^{\text{th}}$  category

$n \times p_j$  the expected number for the  $j^{\text{th}}$  category (with  $p_j$  is the expected proportion of  $X_i$ 's falling in the  $j$ th interval)

with  $p_j = \int_{a_{j-1}}^{a_j} \hat{f}(x) dx$  (continuous fct) or  $p_j = \sum_{a_{j-1} \leq x < a_j} \hat{p}(x)$  (discrete)

# Theoretical distributions

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- Step 3: Determining how representative the fitted distributions are

- Goodness-of-Fit tests: Chi-square test

- Test criterion:  $X^2 = \sum_{j=1}^k \frac{(n_j - np_j)^2}{np_j}$  with  $n = \sum_{j=1}^k n_j$

$X^2$  approximately  $\chi^2$  distributed with  $k-1$  degrees of freedom

$H_0$  hypothesis accepted at level  $\alpha$  if  $X^2 < \chi_{k-1, 1-\alpha}^2$

=> If this value is too large: the observed counts are too different from the expected counts

- Remark:
  - How many intervals? We should have at least five samples in each interval, i.e.  $np_j \geq 5$
  - Where to place the intervals? Determine cutoff values such that we expect an equal number of samples in each interval (= 100%/(# intervals))

# Theoretical distributions

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- Step 3: Determining how representative the fitted distributions are

- Goodness-of-Fit tests: Chi-square test

- Example: Example: 237 random observations of an exponential distribution ( $\lambda = 1$ )

Fitted exponential distribution:  $\hat{f}(x) = \lambda e^{-\lambda x}$  or  $\hat{F}(x) = 1 - e^{-\lambda x}$  with  $\lambda = 0.9722$

Form  $k = 20$  intervals with  $p_j = \frac{1}{k} = 0.05$  for  $j = 1, 2, \dots, 20$  then  $np_j = 237 \times 0.05 = 11.85$

Cut-off values intervals?  $a_0 = 0$  and  $a_{20} = +\infty$

For  $j = 1, 2, \dots, 19$   $\hat{F}(a_j) = \frac{j}{20}$  or  $a_j = -\frac{1}{\lambda} \ln(1 - \frac{j}{20})$



# Theoretical distributions

- Step 3: Determining how representative the fitted distributions are

➤ Goodness-of-Fit tests: Chi-square test

- Example: Example: 237 random observations of an exponential distribution ( $\lambda = 1$ )

| $j$ | Interval  |           | $n_j$ | $np_j$ | $\frac{(n_j - np_j)^2}{np_j}$ |
|-----|-----------|-----------|-------|--------|-------------------------------|
|     | $a_{j-1}$ | $a_j$     |       |        | $np_j$                        |
| 1   | 0.000     | 0.053     | 11    | 11.85  | 0.061                         |
| 2   | 0.053     | 0.108     | 19    | 11.85  | 4.314                         |
| 3   | 0.108     | 0.167     | 10    | 11.85  | 0.289                         |
| 4   | 0.167     | 0.230     | 15    | 11.85  | 0.837                         |
| 5   | 0.230     | 0.296     | 11    | 11.85  | 0.061                         |
| 6   | 0.296     | 0.367     | 12    | 11.85  | 0.002                         |
| 7   | 0.367     | 0.443     | 14    | 11.85  | 0.390                         |
| 8   | 0.443     | 0.525     | 13    | 11.85  | 0.112                         |
| 9   | 0.525     | 0.615     | 10    | 11.85  | 0.289                         |
| 10  | 0.615     | 0.713     | 11    | 11.85  | 0.061                         |
| 11  | 0.713     | 0.821     | 14    | 11.85  | 0.390                         |
| 12  | 0.821     | 0.942     | 14    | 11.85  | 0.390                         |
| 13  | 0.942     | 1.080     | 7     | 11.85  | 1.985                         |
| 14  | 1.080     | 1.238     | 13    | 11.85  | 0.112                         |
| 15  | 1.238     | 1.426     | 4     | 11.85  | 5.200                         |
| 16  | 1.426     | 1.655     | 12    | 11.85  | 0.002                         |
| 17  | 1.655     | 1.951     | 10    | 11.85  | 0.289                         |
| 18  | 1.951     | 2.368     | 8     | 11.85  | 1.251                         |
| 19  | 2.368     | 3.081     | 14    | 11.85  | 0.390                         |
| 20  | 3.081     | $+\infty$ | 15    | 11.85  | 0.837                         |

$$\chi^2 = \sum_{j=1}^k \frac{(n_j - np_j)^2}{np_j} = 17.262$$

$$\chi_{k-1, 1-\alpha}^2 = 30.14 \quad \text{with } k = 20 \text{ and } \alpha = 0.05$$

$$\Rightarrow \chi^2 < \chi_{19, 0.95}^2 \quad \text{Accept } H_0$$

# Theoretical distributions

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- Step 3: Determining how representative the fitted distributions are

## ➤ Goodness-of-Fit tests

### ▪ Kolmogorov-Smirnov test

- Take a large number of samples  $n$ . For each sample  $x_i$  we can compare the proportion of samples less than  $x_i$  to the proportion of samples we would expect to have seen. Compare the empirical distribution function with the theoretical distribution function:

$$\text{Empirical distribution: } F_n(x) = \frac{\text{the number of } x_i \text{ values } \leq x}{n} \quad (\text{step function})$$

Fitted distribution  $\hat{F}(x)$  is the theoretical probability returning a value  $\leq x$

- Compare both functions via the Kolmogorov-Smirnov test and calculate

$$D_n^+ = \max_{\infty} (F_n(x) - \hat{F}(x)) = \max_{1 \leq i \leq n} \left( \frac{i}{n} - \hat{F}(X_{(i)}) \right) \quad \text{the maximum value by which } F_n(x) > \hat{F}(x)$$

$$D_n^- = \max_{\infty} (\hat{F}(x) - F_n(x)) = \max_{1 \leq i \leq n} \left( \hat{F}(X_{(i)}) - \frac{i-1}{n} \right) \quad \text{the maximum value by which } F_n(x) < \hat{F}(x)$$

=> The K-S statistic is simply the largest distance between  $F_n(x)$  and  $\hat{F}(x)$  for all values of  $x$ , i.e.  $D_n = \max\{D_n^+, D_n^-\}$

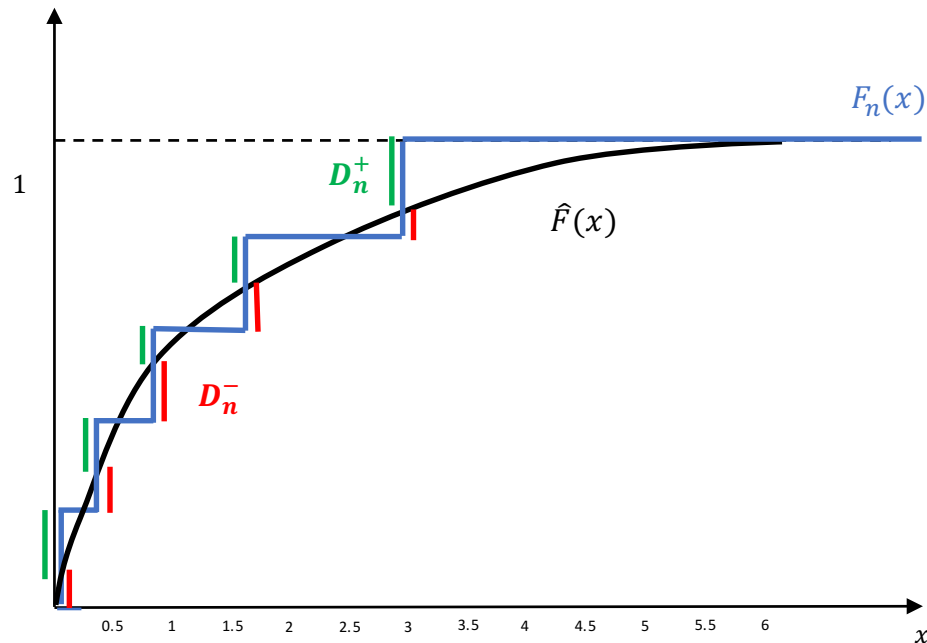
# Theoretical distributions

- Step 3: Determining how representative the fitted distributions are

- Goodness-of-Fit tests

- Kolmogorov-Smirnov test

(not in your slides)



# Theoretical distributions

- Step 3: Determining how representative the fitted distributions are

## ➤ Goodness-of-Fit tests

### ▪ Kolmogorov-Smirnov test

- K-S test:  $D_n$  should lie below threshold value given confidence level  $1 - \alpha$  and tested distribution

| Case                    | Adjusted test statistic                                         | 1 - $\alpha$ |       |       |       |       |
|-------------------------|-----------------------------------------------------------------|--------------|-------|-------|-------|-------|
|                         |                                                                 | 0.85         | 0.9   | 0.95  | 0.975 | 0.99  |
| All parameters known    | $(\sqrt{n} + 0.12 + \frac{0.11}{\sqrt{n}})D_n$                  | 1.138        | 1.224 | 1.358 | 1.480 | 1.628 |
| $N(\bar{X}(n), S^2(n))$ | $(\sqrt{n} - 0.01 + \frac{0.85}{\sqrt{n}})D_n$                  | 0.775        | 0.819 | 0.895 | 0.955 | 1.035 |
| $expo(\bar{X}(n))$      | $(\sqrt{n} + 0.26 + \frac{0.5}{\sqrt{n}})(D_n - \frac{0.2}{n})$ | 0.926        | 0.990 | 1.094 | 1.190 | 1.308 |

- We reject if K-S test statistic  $> c_{1-\alpha}$

# Theoretical distributions

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- Step 3: Determining how representative the fitted distributions are

- Goodness-of-Fit tests

- Example: Example: 237 random observations of an exponential distribution ( $\lambda = 1$ )

$$\hat{F}(X_{(i)}) = 1 - e^{-\lambda X_{(i)}}$$

Calculations cf. Excel file ('Selecting input probability distributions – Calculations.xls')

$$D_n^+ = \max_{1 \leq i \leq n} \left( \frac{i}{n} - \hat{F}(X_{(i)}) \right) = 0.0537$$

$$D_n^- = \max_{\infty} \left( \hat{F}(X_{(i)}) - \frac{i-1}{n} \right) = 0.0268$$

$$\Rightarrow D_n = 0.0537$$

$$\Rightarrow \text{Test statistic exponential distribution: } \left( \sqrt{n} + 0.26 + \frac{0.5}{\sqrt{n}} \right) \left( D_n - \frac{0.2}{n} \right) = 0.815$$

K-S test:  $0.815 < \text{Threshold value } c_{1-\alpha}(\text{exponential distribution, } 1-\alpha = 0.95) = 1.094$

$\Rightarrow \text{Accept } H_0$